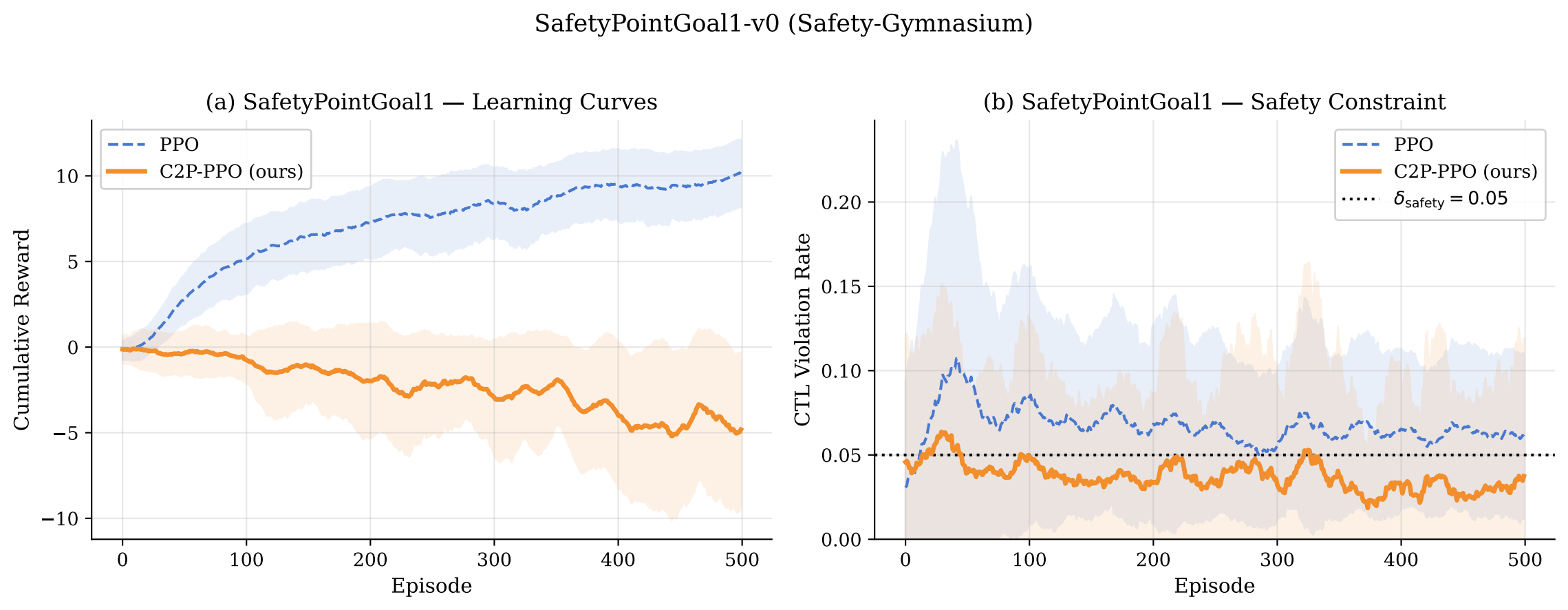

SafetyPointGoal1-v0 (Safety-Gymnasium)
(a) SafetyPointGoal1 — Learning Curves
PPO
C2P-PPO (ours)
Cumulative Reward
10
5
0
−5
−10
0
100
200
300
400
500
Episode
(b) SafetyPointGoal1 — Safety Constraint
PPO
C2P-PPO (ours)
$\delta_{\text{safety}} = 0.05$
CTL Violation Rate
0.20
0.15
0.10
0.05
0.00
0
100
200
300
400
500
Episode